

Exp No:28 COUNTING SEMAPHORE IN SYSTEMC

DATE: 25/08/2026

```
kaviya@KV: ~
GNU nano 8.7.1
#include <systemc.h>

SC_MODULE(counting_semaphore) {

    int sem;

    void task() {
        while(true) {
            wait(1, SC_SEC);

            if(sem > 0) {
                sem--;
                cout << "Task Accessing Resource at "
                << sc_time_stamp()
                << " Remaining: " << sem << endl;

                wait(2, SC_SEC);
                sem++;
            }
        }
    }

    SC_CTOR(counting_semaphore) {
        sem = 2;
        SC_THREAD(task);
        SC_THREAD(task);
        SC_THREAD(task);
    }
};

int sc_main(int argc, char* argv[]) {
    counting_semaphore obj("Counting");
    sc_start(10, SC_SEC);
    return 0;
}
```

```
kaviya@KV:~$ nano exp28.cpp
kaviya@KV:~$ g++ -o exp28 exp28.cpp -lsystemc
kaviya@KV:~$ ./exp28
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
```

```
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: kernel/sc_object_manager.cpp:153
```

```
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: kernel/sc_object_manager.cpp:153
```

```
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
```

```
kaviya@KV:~$ |
```